

ALEXANDER TOBIN

(757) 289-1204 ♦ atobin@alum.utk.edu ♦ Virginia Beach, VA

CORE SKILLS

Databricks · Delta Live Tables
· Unity Catalog

Generative AI · LLMs · RAG
Architecture

Databricks AI Genie · MLflow ·
MLOps

PySpark · Spark SQL · Spark
Optimization

Medallion Architecture · Delta
Lake

Python · AI Workflow Automation

AWS · Azure · GCP

CI/CD Pipelines · Git · GitHub
Actions

Power BI · SQL Optimization

Technical Leadership · Engineer
Mentorship

Stakeholder Management · Agile
/ JIRA

EDUCATION

Bachelor of Arts — Pre Law/Legal
Studies, Business Concentration
University of Tennessee, Knoxville,
TN — Aug 2019

CERTIFICATIONS

- ♦ DP-203: Azure Data Engineering (In Progress)
- ♦ Google Project Management Certification
- ♦ Google Analytics Certification

PROFESSIONAL SUMMARY

Senior AI & Data Engineering leader with 7+ years of hands-on experience designing and deploying enterprise-scale AI, LLM, and Lakehouse solutions on Databricks. Deep expertise in Generative AI, RAG architectures, LLM workflow integration, PySpark, Spark SQL, Medallion Architecture, Delta Live Tables, Unity Catalog, and MLOps across AWS, Azure, and GCP. Proven technical leader who mentors engineers, shapes AI architecture, and contributes code — 70% hands-on development, strong stakeholder communication, and a track record of delivering high-impact AI systems into production. DP-203 Azure Data Engineering in progress.

PROFESSIONAL EXPERIENCE

Freelance AI Tech Lead & Senior Data Engineer

Jan 2026 – Present

Self-Employed

- ♦ [callcathai.online](#) — Designed and deployed Generative AI and LLM-integrated call analytics platform on Databricks; built Medallion Architecture pipelines with Delta Live Tables, RAG-based insight retrieval, and AI-assisted executive reporting with automated alerting and monitoring.
- ♦ [primelayertrades.com](#) — Architected enterprise AI platform on Databricks with LLM workflow integration, Unity Catalog governance, MLflow experiment tracking, and CI/CD pipelines for production AI model deployment across GCP and AWS environments.
- ♦ [hereraspreadsheets.com](#) — Built reusable AI and data engineering frameworks including Spark optimization patterns, Generative AI workflow automation, and streaming data pipelines feeding real-time executive dashboards; mentored client engineering team on MLOps and AI development best practices.
- ♦ Led AI architecture design and technical stakeholder engagement for client organizations; delivered AI-powered business intelligence solutions with C-suite visibility and measurable operational impact.

Operations Manager / AI & Data Engineering Lead

Nov 2024 – Jan 2026

Catalyzt Nutrition, Virginia Beach, VA

- ♦ Designed and led delivery of Databricks Lakehouse platform on GCP; implemented Medallion Architecture, Delta Live Tables, and Delta Lake with AI-integrated analytics using Databricks AI Genie for executive decision support.
- ♦ Developed PySpark ETL/ELT pipelines and Spark SQL transformations; integrated LLM-powered analysis workflows and Generative AI tooling into production reporting pipelines for operational and financial analytics.
- ♦ Implemented MLOps practices including CI/CD pipeline deployment, model tracking with MLflow, and Unity Catalog governance controls across all AI and data engineering environments.
- ♦ Mentored junior engineers on Python, PySpark, and AI development best practices; led technical direction and cross-functional stakeholder engagement across the technology organization.

Data Analyst / Project Manager

Mar 2022 – Jul 2024

Ferguson Enterprises Inc., Remote

- ♦ Designed and built enterprise-scale Databricks Lakehouse on Azure and GCP; implemented Medallion Architecture pipelines with PySpark and Spark SQL processing large ecommerce and supply chain datasets.

Supply Chain Data Analyst

Jan 2021 – Dec 2021

ZenEvo Chocolate, Knoxville, TN

Data Analyst Student Assistant

Nov 2014 – Dec 2019

University of Tennessee